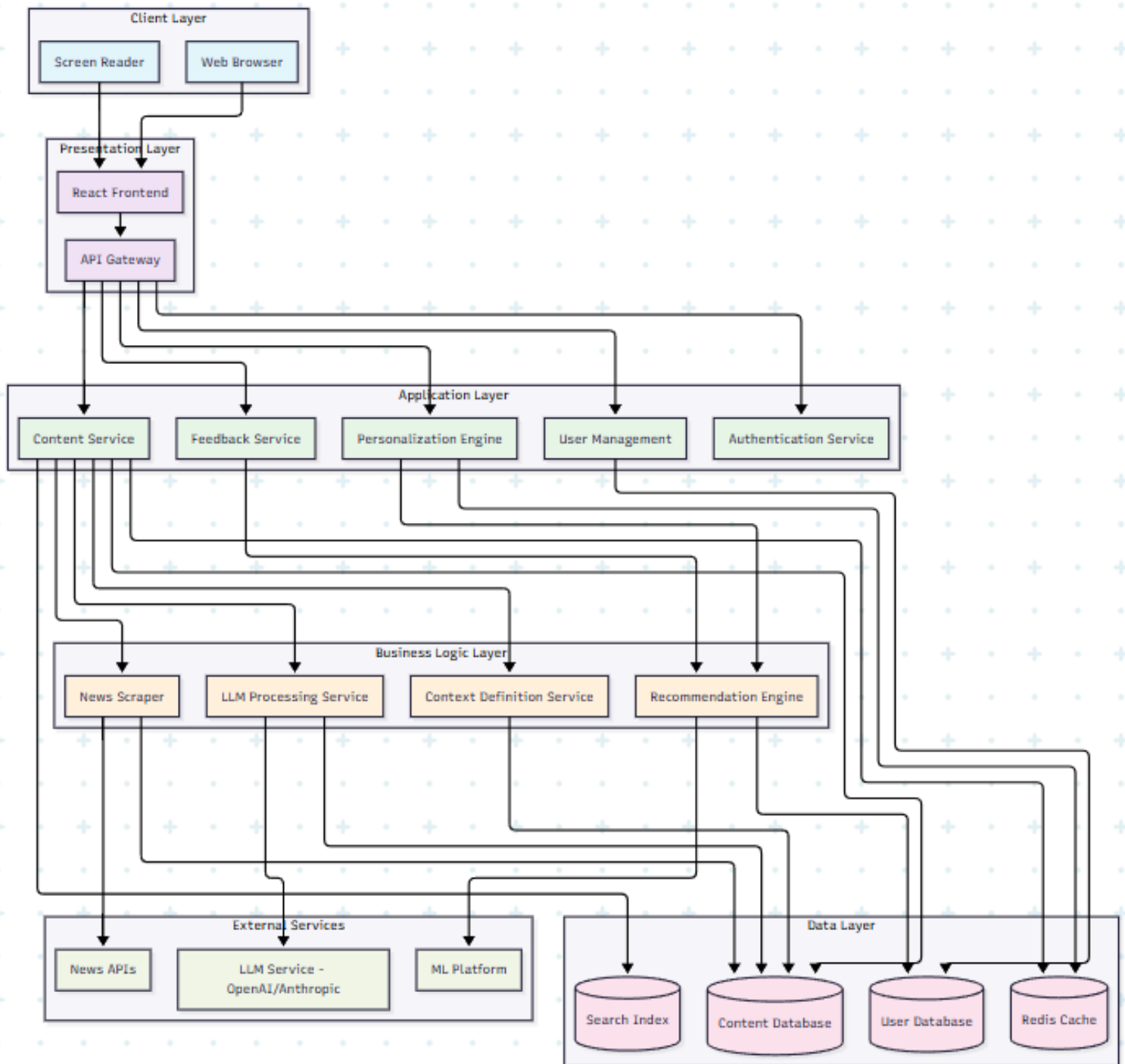


# High-Level Architecture UML Diagram



# 1. Architecture Style Rationale

We have decided on a **Layered Architecture with Service-Oriented Components** for this project:

- **Separation of Concerns:** Clear distinction between presentation, business logic, data access, and external integrations
- **Scalability:** Individual layers can be optimized and scaled independently
- **Maintainability:** Well-defined boundaries make the system easier to modify and extend
- **Testability:** Each layer can be unit tested in isolation
- **Flexibility:** Supports future migration to microservices if needed without major restructuring

This approach balances simplicity for initial development while providing a foundation for growth.

# 2. Technical Constraints Summary

## Programming Languages & Frameworks

### Frontend:

- **React.js** with **TypeScript** for type safety and maintainability
- **Next.js** for server-side rendering and SEO optimization
- **Tailwind CSS** for responsive, accessible styling
- **React Aria** for comprehensive screen reader support

### Backend:

- **Node.js** with **Express.js** or **Fastify** for API services
- **Python** for ML/AI processing services (recommendation engine, LLM integration)
- **TypeScript** for consistent type checking across JavaScript services

### Database & Storage:

- **PostgreSQL** for structured user and content data
- **Redis** for caching and session management
- **Elasticsearch** for full-text search and content indexing

## Key APIs & External Services

### News Aggregation:

- **NewsAPI**, **Guardian API**, **Reddit API** for content sources
- **RSS feed parsers** for additional news sources

### AI/ML Services:

- **OpenAI GPT-4** or **Anthropic Claude** for content simplification
- **Hugging Face** for additional NLP tasks
- **TensorFlow Serving** or **AWS SageMaker** for recommendation models

**Infrastructure:**

- **AWS** or **Google Cloud** for hosting and managed services
- **Docker** for containerization
- **GitHub Actions** for CI/CD

**Accessibility Requirements**

- WCAG 2.1 AA compliance
- Semantic HTML structure
- ARIA labels and roles
- Keyboard navigation support
- High contrast mode compatibility